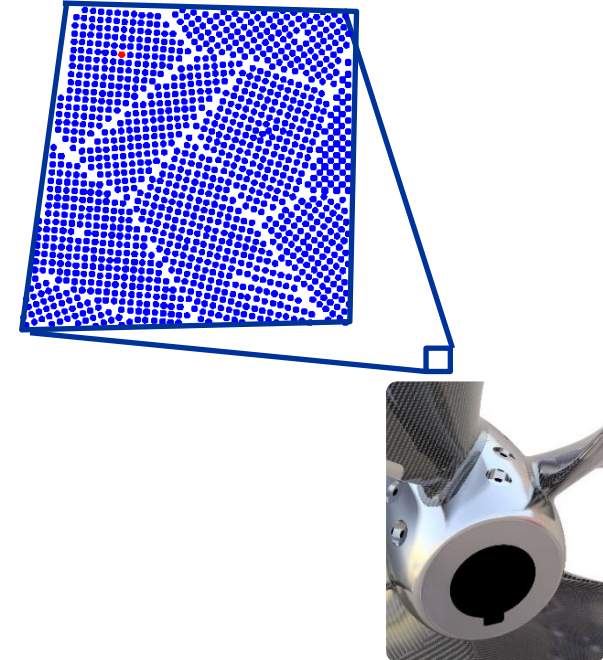


TOPIC 2. STRUCTURE OF MATERIALS II

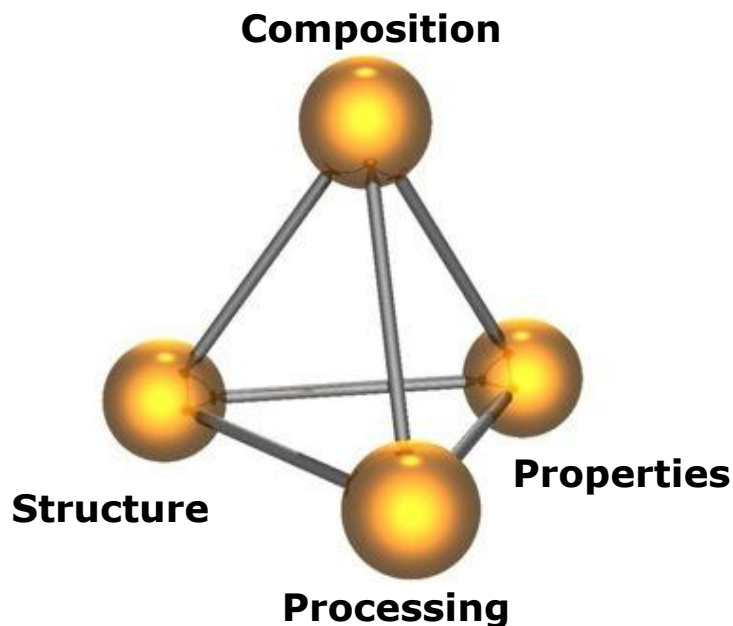
- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- Solid solutions
- Crystalline defects: linear defects
- Crystalline defects: planar defects
- Polymorphism and Allotropy



IN PREVIOUS EPISODES...

Materials Science and Engineering

- Materials science provides the “toolbox” for the rest of technologies
- Without continuous progress in materials, most other technological areas would stagnate (Electronics, Energy storage, Medicine, Aerospace, etc.).



- **Composition:** the **chemical** make-up of a material or what materials are made of.
- **Structure:** the **arrangement of atoms** in the material on a given scale
- **Properties:** how are **behave** in different conditions and environments
- **Processing:** how are materials **shaped** into useful forms

IN PREVIOUS EPISODES...

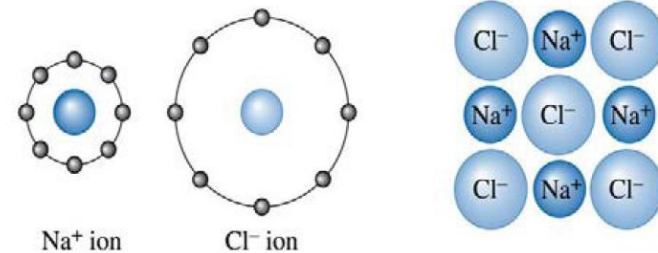
Atomic bonding

- It involves the transfer or sharing of electrons between atoms, resulting in electrostatic or mutual attractions

- 3 types:

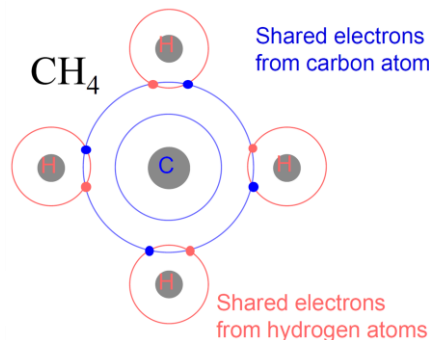
- Ionic:**

- Electrons transfer between + and – ions
- Need $\uparrow \Delta\chi$ (electronegativity difference)



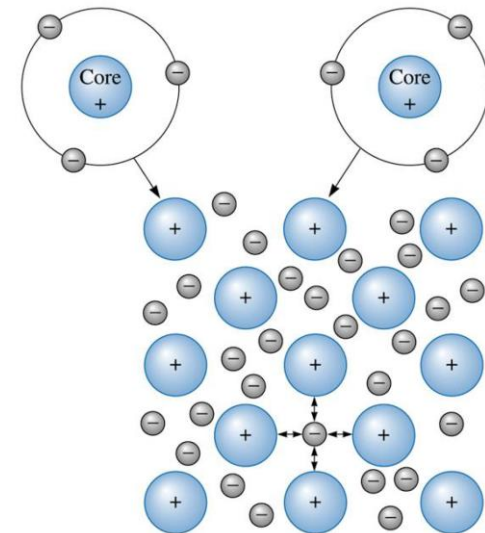
- Covalent:**

- Share electrons
- Need similar χ ($\downarrow \Delta\chi$)



- Metallic:**

- e⁻ sea formed by donation of valence electrons
- Metals and alloys

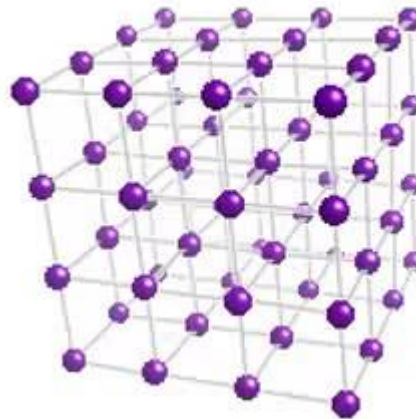


IN PREVIOUS EPISODES...

Crystalline structures



How are atoms arranged in a Golden World Cup? (Or in a metallic material)



https://www.doitpoms.ac.uk/tlplib/miller_indices/lattice_index.php

Crystalline Lattice

Unit Cell

IN PREVIOUS EPISODES...

Crystalline structures

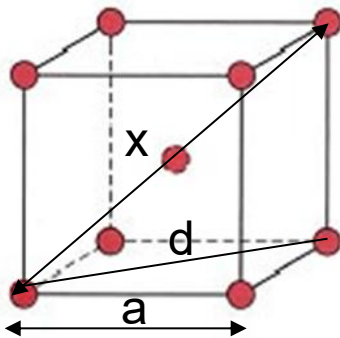
- 14 Bravais lattices define every possible unit cell
- **3 most important** (90% metals follow them):

- N° of tetrahedral sites = $2n$
- N° of octahedral sites = n

BODY-CENTERED CUBIC (BCC)

FACE-CENTERED CUBIC (FCC)

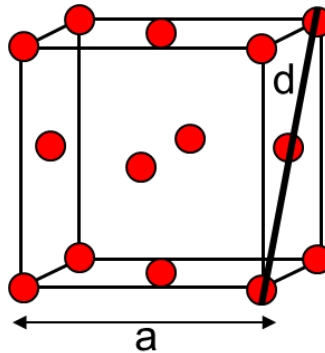
HEXAGONAL CLOSE-PACKED (HCP)



$$4R = \sqrt{3}a$$

$$n_{atoms} = 1 + 8 \cdot \frac{1}{8} = 2at$$

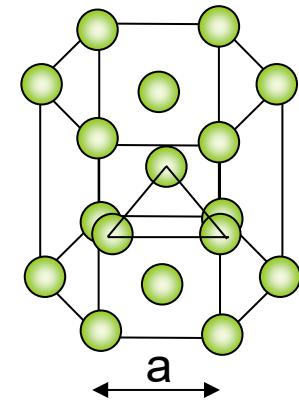
$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi R^3 \right] \cdot 2}{\left[\frac{4R}{\sqrt{3}} \right]^3} = \frac{\sqrt{3}}{8} \pi = 0.68$$



$$4R = \sqrt{2}a$$

$$n_{atoms} = 8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4at$$

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi R^3 \right] \cdot 4}{\left[2\sqrt{2}R \right]^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$



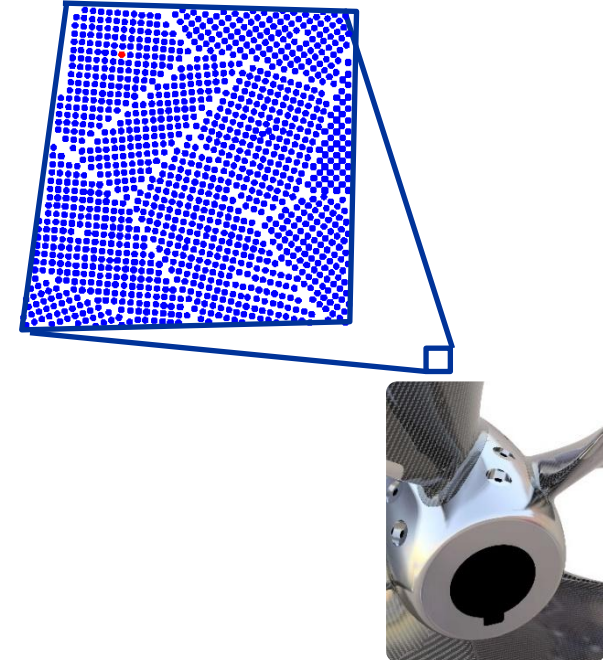
$$2R = a$$

$$n_{atoms} = \left[12 \cdot \frac{1}{6} \right] + \left[2 \cdot \frac{1}{2} \right] + 3 = 6at$$

$$APF = \frac{V_{atoms}}{V_{cell}} = \frac{\left[\frac{4}{3} \pi R^3 \right] \cdot 6}{24 \cdot R^3 \sqrt{2}} = 0.74$$

TOPIC 2. STRUCTURE OF MATERIALS II

- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- Solid solutions
- Crystalline defects: linear defects
- Crystalline defects: planar defects
- Polymorphism and Allotropy



ATOMIC DENSITY IN CRYSTALS

Volume density

$$\rho_V = \frac{\text{mass of unit cell}}{\text{volume of unit cell}} = \frac{n^\circ \text{ atoms per cell} \cdot \text{mass of atom}}{\text{volume of unit cell}}$$

$$\rho = \frac{n \cdot M}{V_{\text{Cell}} N_A}$$

Example: Cu in a FCC

Data: $M(\text{Cu}) = 63.5 \text{ g/mol}$; $R = 1.28 \text{ \AA}$

$$n = 8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4 \text{ atoms}$$

$$V_C = a^3 = (2R\sqrt{2})^3 = 16\sqrt{2}R^3$$

$$\rho_V = \frac{n^\circ \text{ at}_{\text{celda}}}{V_{\text{celda}}} = \frac{4 \text{ at}}{(2\sqrt{2} \cdot 1.28 \cdot 10^{-8} \text{ cm})^3} = 8.43 \cdot 10^{22} \frac{\text{at}}{\text{cm}^3}$$

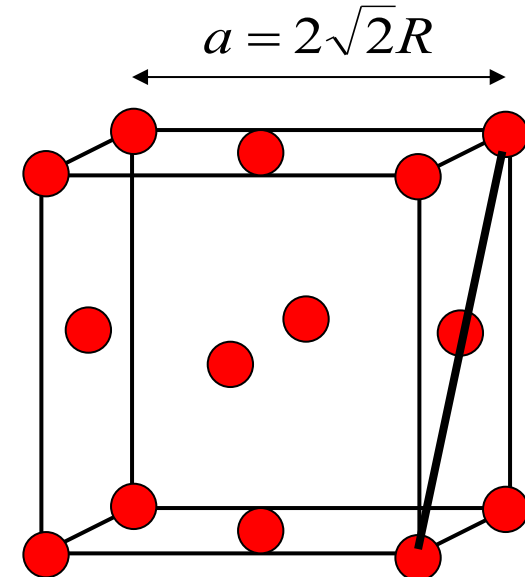
$$\rho = \frac{4 \text{ atoms} \cdot 63.5 \frac{\text{g}}{\text{mol}}}{6.023 \cdot 10^{23} \frac{\text{atoms}}{\text{mol}} \cdot (2\sqrt{2} \cdot 1.28 \cdot 10^{-8} \text{ cm})^3} = 8.89 \frac{\text{g}}{\text{cm}^3}$$

n : number of atoms /unit cell

M : atomic weight

V_C : volume of the unit cell

N_A : Avogadro number ($6.023 \cdot 10^{23} \text{ atoms/mol}$)



8.94 g/cm³ from the bibliography

ATOMIC DENSITY IN CRYSTALS

Linear density

$$\rho_l = \frac{\text{N}^\circ \text{ atoms intersected diametrically by the line}}{\text{length of line}} = \frac{\text{atoms}}{m}$$

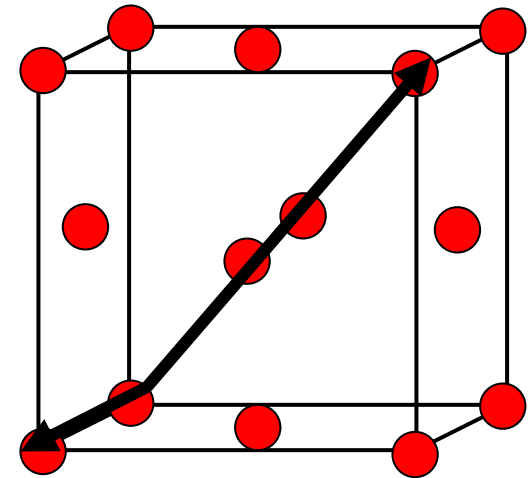
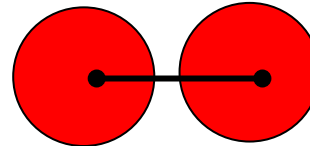
$$\rho_{[uvw]} = \frac{n^\circ at_{[uvw]}}{L_{[uvw]}}$$

Example: Cu in FCC, direction: [100]

Datas: $M(\text{Cu}) = 63.5 \text{ g/mol}$; $R = 1.28 \text{ \AA}$

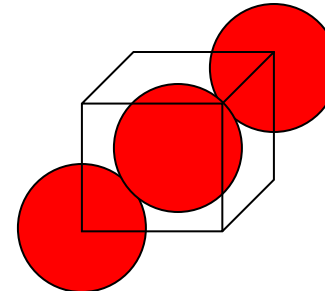
$$n^\circ \text{ atoms} = 2 \times \frac{1}{2} \text{ vertices} = 1 \text{ atom}$$

$$\rho_{[100]} = \frac{1 \text{ atom}}{2\sqrt{2} \cdot 1.28 \cdot 10^{-8} \text{ cm}} = 2.76 \cdot 10^7 \frac{\text{atom}}{\text{cm}}$$



Example: Cu in FCC, direction: [111]

$$\rho_{[111]} = \frac{1 \text{ central atom} + 2 \cdot \frac{1}{2} \text{ vertices}}{\sqrt{3} \cdot a} = \frac{2\sqrt{3}}{3a} \frac{\text{atoms}}{m}$$



ATOMIC DENSITY IN CRYSTALS

Planar density

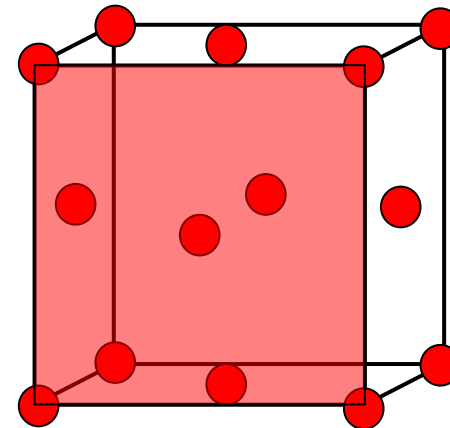
$$\rho_p = \frac{\text{N}^\circ \text{ atoms intersected by the area/plane}}{\text{area of plane}} = \frac{\text{atomss}}{m^2}$$

$$\rho_{(hkl)} = \frac{n^\circ at_{(hkl)}}{A_{(hkl)}}$$

Example: Cu in FCC, plane: (100)

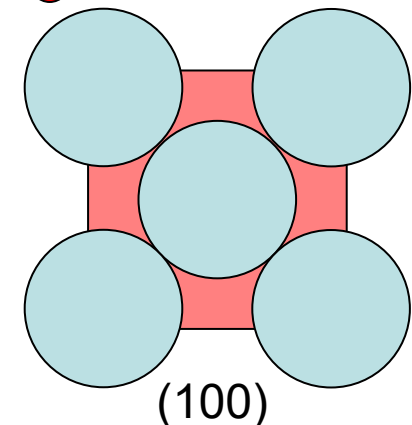
Data: $M(\text{Cu}) = 63.5 \text{ g/mol}$; $R = 1.28 \text{ \AA}$

$$a = 2\sqrt{2}R$$

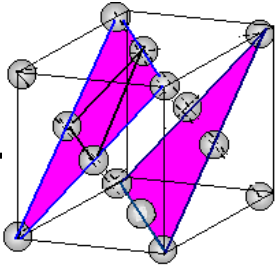
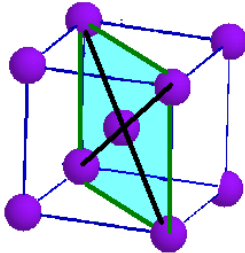
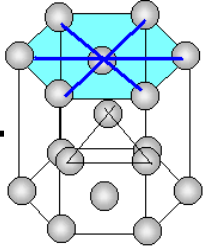


$$\rho_{(100)} = \frac{1 \text{ central atom} + 4 \cdot \frac{1}{4} \text{ vertices}}{a^2} = \frac{2 \text{ at}}{a^2} = \frac{2 \text{ at}}{(2\sqrt{2} \cdot 1.28 \cdot 10^{-8} \text{ cm})^2}$$

$$\rho_{(hkl)} = 1.5 \cdot 10^{15} \frac{\text{at}}{\text{cm}^2}$$

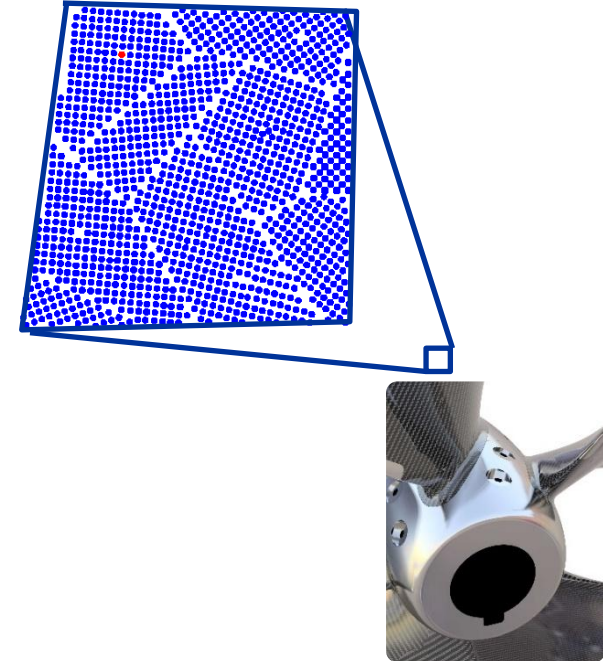


Summary

Type of lattice	FCC	BCC	HCP
N ^o atoms/cell	$\bullet 8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4$	$\bullet 8 \cdot \frac{1}{8} + 1 = 2$	$\bullet 12 \cdot \frac{1}{6} + 2 \cdot \frac{1}{2} + 3 = 6$
Atomic Radius	$r = \frac{a}{2\sqrt{2}}$	$r = \frac{a\sqrt{3}}{4}$	$r = \frac{\sqrt{\frac{a^2}{3} + \frac{c^2}{4}}}{2}$
V_{cell} and V_{interstices}	$\mathbf{V_{cell} = a^3}$ $V_h = V_c - 4\left(\frac{4}{3}\pi r^3\right) = 5,872r^3$	$\mathbf{V_{cell} = a^3}$ $V_h = V_c - 2\left(\frac{4}{3}\pi r^3\right) = 3,94r^3$	$\mathbf{V_{cell} = a^2 \cdot c \cdot \frac{3\sqrt{3}}{2}}$ $V_h = V_c - 6\left(\frac{4}{3}\pi r^3\right) = 8,808r^3$
Volume relative of interstitials	$V_r = \frac{V_h}{V_c} = 0,26$	$\mathbf{V_r = 0,32}$	$\mathbf{V_r = 0,26}$
Octahedral Interstitials	N ^o =4 R=1/2(a-2r)=0,41r	N ^o =6 R=1/2(a-2r)=0,15r	N ^o =6 R=0,41r
Tetrahedral Interstitials	N ^o =8 $R_t = \left[\sqrt{\frac{3}{2}} - 1 \right] r = 0,225r$	N ^o =12 $R_t = \left[\sqrt{\frac{5}{3}} - 1 \right] r = 0,291r$	N ^o =12 R _t =0,225r
Close packed planes	{111} there are 4	{110} there are 6	{0001} there is 1
Closed packed directions	<110> there are 6 	<111> there are 4 	<2110> there are 3 -- 

TOPIC 2. STRUCTURE OF MATERIALS II

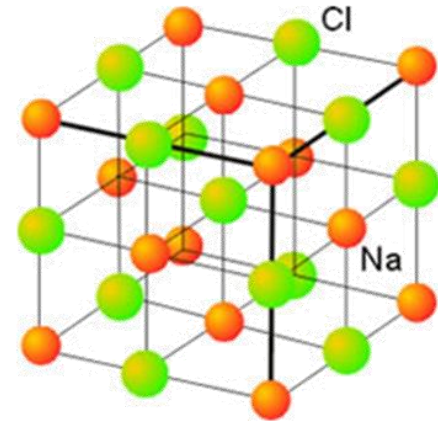
- Calculation of the atomic, linear, planar density
- **Crystalline defects: point defects**
- Solid solutions
- Crystalline defects: linear defects
- Crystalline defects: planar defects
- Polymorphism and Allotropy



PERFECT AND IMPERFECT CRYSTALS

- **Perfect crystal**

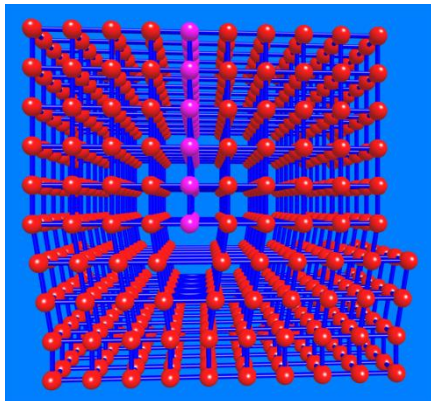
- All atoms in the regular lattice positions for a given structure
- All distances d_{hkl} are the same for the given (h k l)
- No missing or extra atoms
- Only possible at 0 K



Perfect crystal (NaCl)

- **Real crystal**

- Atoms vibrate
- There are positions that are not occupied (vacancies)
- Atoms displaced from ideal positions
- There are defects that modify the properties



“Crystalline defect”: Disruption in the perfect periodic arrangement of atoms/ions in a crystalline material.

PERFECT AND IMPERFECT CRYSTALS

Thermodynamics $\Rightarrow$ Justify the presence of defects

Creation of a defect $\Rightarrow \uparrow \Delta H, \uparrow \Delta S \Rightarrow \downarrow \Delta G_{\text{crystal}}$

Vacancy or defect:

$$\Delta G = \Delta H - T\Delta S < 0$$

$$[\text{Defects}]_{\text{eq}} \Rightarrow \Delta G = 0$$

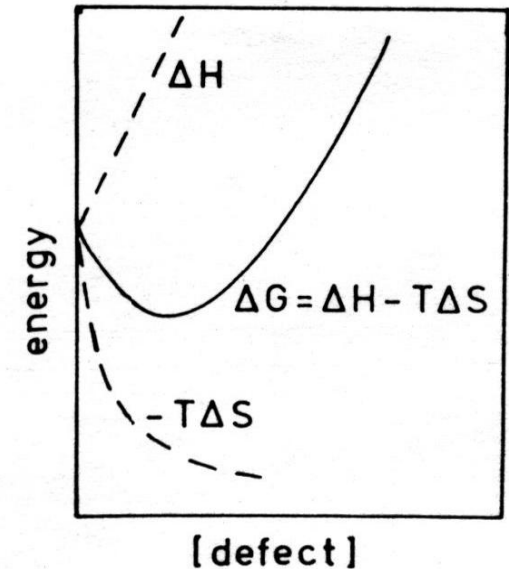
$$\Delta H < T\Delta S$$

ΔG : Gibbs energy

ΔH : enthalpy of formation of each vacancy

ΔS : entropy of formation of each vacancy

T: absolute temperature



Configuration Entropy $\Rightarrow S = K_B \ln W$

K_B = Boltzmann constant = $1.380622 \cdot 10^{-23} \text{ JK}^{-1}$

W = Way of distributing n defects in N possible random positions

In 1 mol of C^+ there are N_A possibilities to create 1 vacancy $\Rightarrow W \propto 10^{23}$

$$W = \frac{N!}{(N-n)!n!}$$

When $\uparrow T \Rightarrow \uparrow [\text{defects}]$

DEFECTS TYPES (according to dimensions)**Classification of defects****1. Point**

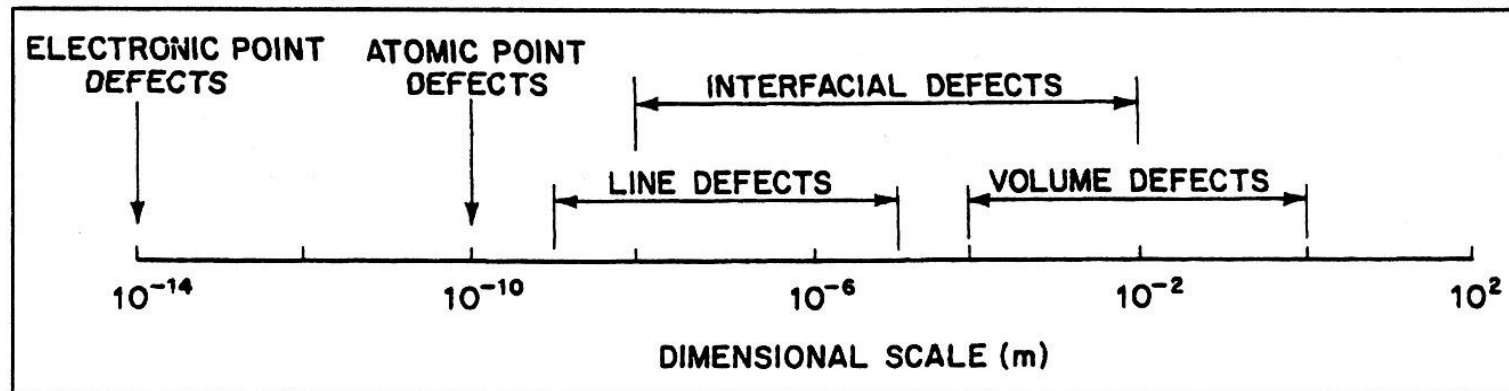
Vacancies
Interstitials
Substitutional
Schottky
Frenkel
Order-Disorder

3. Planar or Extended

External Surfaces
Grain Boundaries
Twin Boundaries
Stacking Faults

2. Linear: dislocations:

Edge (Taylor)
Screw (Burgers)
Mixed

**4. Complex (volume defects):
“clusters”****Figure 4.1** Dimensional ranges of different classes of defects.

POINT DEFECTS: VACANCIES AND INTERSTITIALS

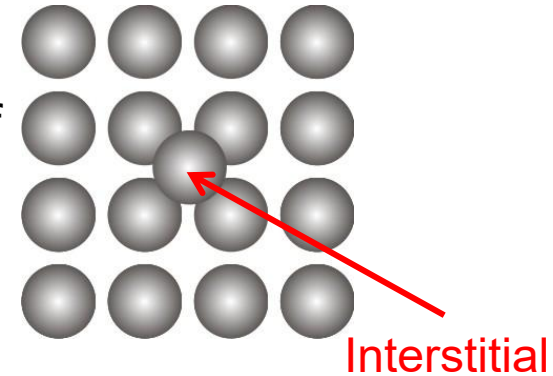
Point Defects: Localized disruptions in an otherwise 'perfect' atomic/ionic arrangement in a crystal structure.

- Although the defect occurs at one or two sites, **their presence is “felt” over much larger distances** in a crystalline material.
- Introduced by the movement of the atoms **when they gain energy (e.g., $\uparrow T$)**.
- Point defects:
 - ✓ **Impurities:** elements/compounds **present from raw materials** or processing.
 - ✓ **Dopants:** elements/compounds **deliberately added** (known concentrations).

POINT DEFECTS: VACANCIES AND INTERSTITIALS

a. Interstitial

- An **atom** from the crystal occupies the place of an (*self-interstitial* or *interstitialcy*)



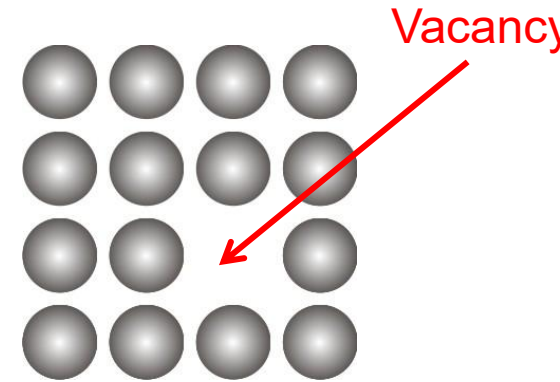
b. Vacancy

- An atom is **missing** from the regular position
- Number of vacancies at temperature T

$$n_v = N \exp\left(\frac{-\Delta H_v}{RT}\right)$$

$$N = \frac{N_A \rho}{M}$$

$n_v = n^\circ$ vacancies per cm^3
 $N = n^\circ$ of lattice points per cm^3
 $\Delta H_v =$ Formation energy (J.mol^{-1})
 $n_v / N =$ fraction of vacancies
 $N_A =$ Avogadro's number
 $M =$ molecular weight
 $\rho =$ density



They are produced:

- During solidification (impurities, alloys)
- By particle bombardment with $\uparrow E$
- During plastic deformation (processing)
- When $\uparrow T \Rightarrow \uparrow$ thermal vibrations $\Rightarrow \uparrow n_v$

For metals, vacancies concentration at equilibrium

$$\frac{n_v}{N} \square 10^{-4} \text{ max}$$

POINT DEFECTS: SCHOTTKY AND FRENKEL

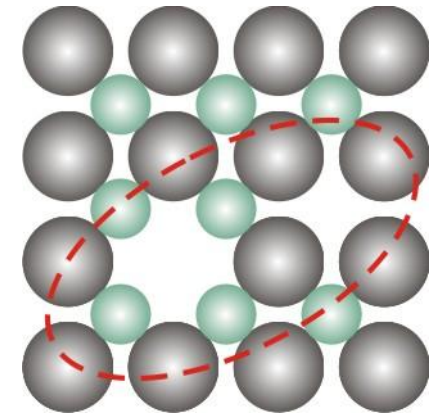
POINT DEFECTS IN IONIC CRYSTALS

c. Schottky defect

- A **pair of vacancies** of opposite charge. In order to maintain neutrality $\Rightarrow$ Anionic and cationic vacancies appear

$$n_s = N \exp\left(-\frac{\Delta H_s}{2RT}\right)$$

$\Delta H_s / \Delta H_F = E$ for Schottky/Frenkel defect creation
 $N = n^\circ$ of lattice positions
 $N_i = n^\circ$ of interstitial positions
 $T =$ Temperature (K)

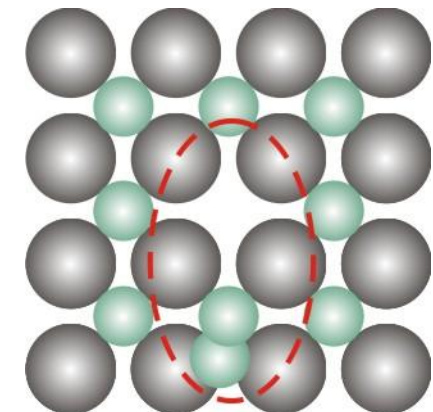


In a crystal $\Delta H_s \neq \Delta H_F \Rightarrow$ the defect with the lowest ΔH will form

d. Frenkel defect

- Migration** of an ion from a regular to an interstitial position (*combination vacancy-interstitial*).
- Number of defects:

$$n_F = \sqrt{NN_i} \exp\left(-\frac{\Delta H_F}{2RT}\right)$$



POINT DEFECTS: SCHOTTKY AND FRENKEL

Entalpía de formación de defectos Schottky y Frenkel en algunos compuestos seleccionados

	Compuesto	$\Delta H(10^{-19} \text{ J})$	$\Delta H(\text{eV})^*$
Defectos Schottky	MgO	10.57	6.60
	CaO	9.77	6.10
	LiF	3.75	2.34
	LiCl	3.40	2.12
	LiBr	2.88	1.80
	LiI	2.08	1.30
	NaCl	3.69	2.30
	KCl	3.62	2.26
Defectos Frenkel	UO ₂	5.45	3.40
	ZrO ₂	6.57	4.10
	CaF ₂	4.49	2.80
	SrF ₂	1.12	0.70
	AgCl	2.56	1.60
	AgBr	1.92	1.20
	β -AgI	1.12	0.70

* La literatura suele citar valores en eV, por lo que éstos se incluyen con fines comparativos:
 $1 \text{ eV} = 1.60219 \times 10^{-19} \text{ J}$.

L. Smart and E. Moore.
 "Química del Estado Sólido".
 Addison-Wesley
 Iberoamericana. Wilmington,
 1995.

Schottky defects

- NaCl** $\Delta H_s = 3.69 \times 10^{-19} \text{ J}$
 $T=300\text{K}$ $n_s = 2.64 \times 10^4 \text{ vacancies/mol}$
 $T=1000\text{K}$ $n_s = 9.38 \times 10^{17} \text{ vacancies/mol}$

- MgO** $\Delta H_s = 10.57 \times 10^{-19} \text{ J}$
 $T=300\text{K}$ $n_s = 2.12 \times 10^{-32} \text{ vacancies/mol}$
 $T=1000\text{K}$ $n_s = 1.39 \times 10^7 \text{ vacancies/mol}$

$n^\circ \text{ defects} \uparrow \text{ with } T$

$\Delta H_s(\text{MgO}) > \Delta H_s(\text{NaCl})$
 n_s low, more difficult to
 create vacancies

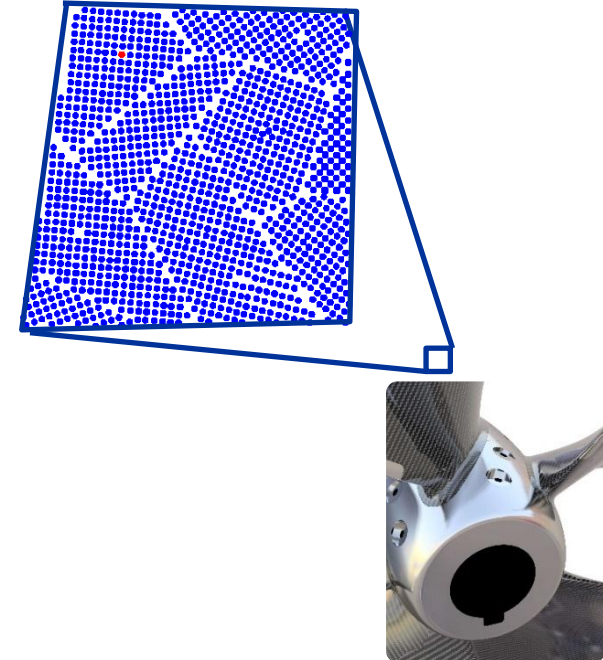
POINT DEFECTS: ORDER-DISORDER IN SOLID SOLUTIONS**e. Order-disorder phenomena** *(in substitutional solid solutions)*

Solids with elements of similar electronegativities.

What is a solid solution?

TOPIC 2. STRUCTURE OF MATERIALS II

- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- **Solid solutions**
- Crystalline defects: linear defects
- Crystalline defects: planar defects
- Polymorphism and Allotropy



SOLID SOLUTION

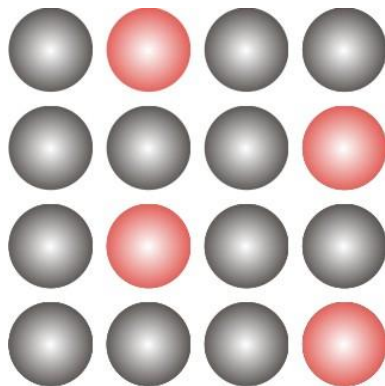
When mixing two different materials in the liquid state and subsequent solidification we can either get:



- **Solid solution:** crystalline solid that has **two** or more **elements dispersed as atoms** in a structure that has **just one phase**. Solid solutions can be:

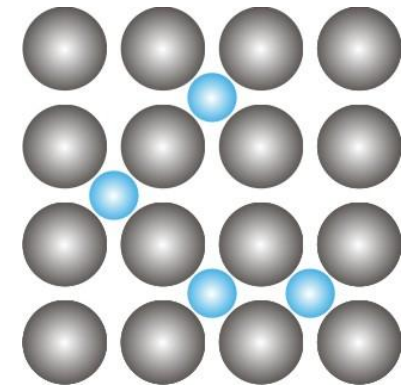
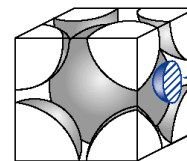
Substitutional solid solutions

Solute atoms replace atoms of the host



Interstitial solid solutions

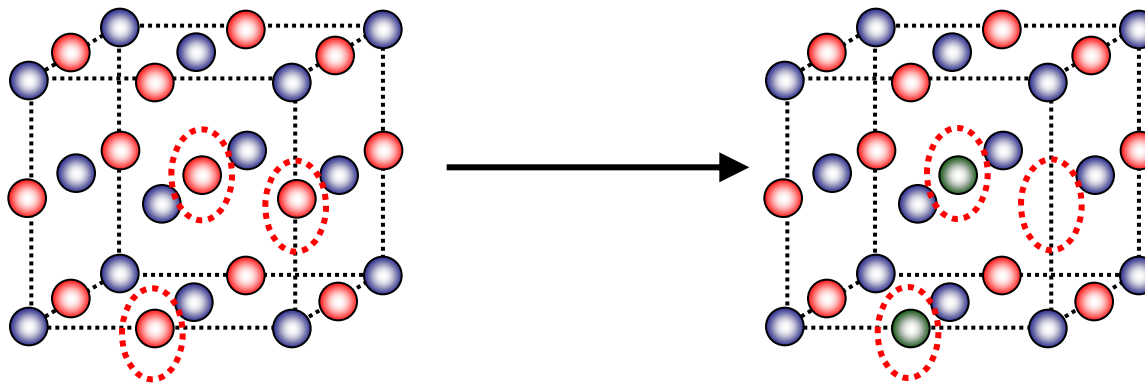
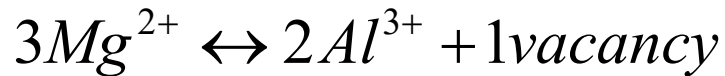
Solute atoms occupy interstitial sites (elements with small radius H, C, N, B, N)



SOLID SOLUTION

Generation of defects due to solid solution in ionic solids

A solid solution of MgO and Al_2O_3 : in a pure MgO crystal Mg atoms are substituted by Al atoms vacancies are created (to maintain electrical neutrality)



SOLID SOLUTION: HUME-ROTHERY RULES

Conditions for the formation of solid solutions Hume-Rothery rules

- **Substitutional solid solutions:**

- The **crystal structure** of solute and solvent must **match**
- Atomic **radii** of solute and solvent atoms must be **similar**
 - For ceramics difference < 30%
 - For metals difference < 15%
 - For differences > 15% limited solubility (<1%)
- Solute and solvent must have **similar electronegativities**
- Solute and solvent must have **the same valence**

- **Interstitial solid solutions:**

- Radii of solute atoms must be **smaller** than those of the solvent
- Solute and solvent must have **similar electronegativities**

If all conditions are fulfilled $\Rightarrow$ we do not necessarily have total solubility

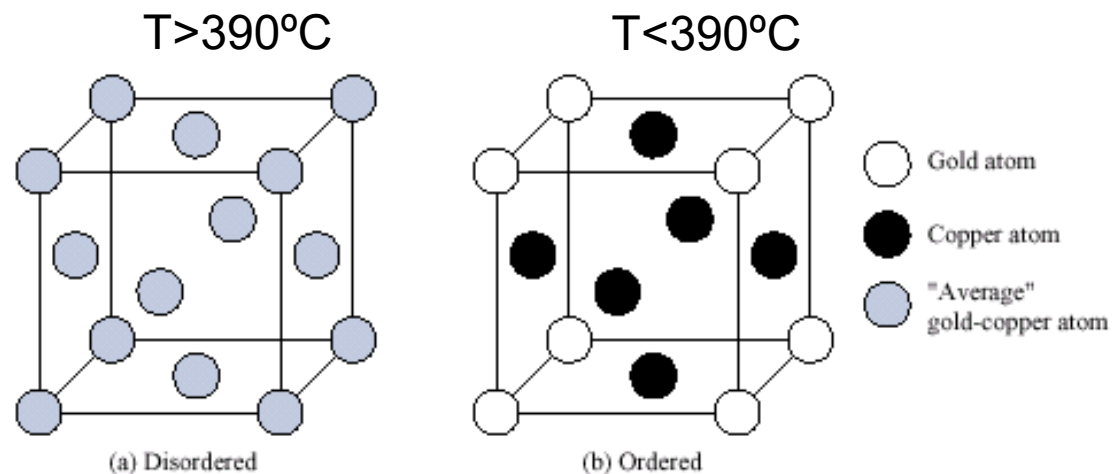
-If one or more conditions are not fulfilled $\Rightarrow$ partial solubility

POINT DEFECTS: ORDER-DISORDER IN SOLID SOLUTIONS**e. Order-disorder phenomena** (*in substitutional solid solutions*)

Solids with elements of similar electronegativities.

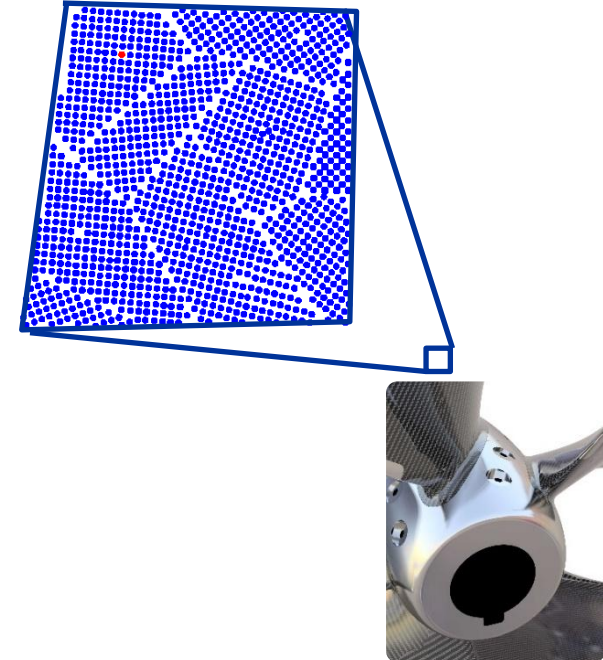
- **Disorder:** Atoms of one sub-lattice occupy positions of the other and vice versa
- **Order:** Above or below certain critical temperature, disordered atoms return to regular positions.

Example: Cu-Au alloy



TOPIC 2. STRUCTURE OF MATERIALS II

- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- Solid solutions
- **Crystalline defects: linear defects**
- Crystalline defects: planar defects
- Polymorphism and Allotropy



LINEAR DEFECTS: DISLOCATIONS

DISLOCATIONS : line imperfections in an otherwise perfect crystal.

Characteristics: they can be displaced in the interior of a crystal by applying relatively low forces and can produce a complete displacement over crystalline planes.

They explain:

- $E_{\text{theoretical}}$ (Young modulus) $> E_{\text{experimental}}$
- **Plastic deformation in metals (workability, ductility)**

Formation:

- during solidification
- upon plastic or permanent deformation
- through concentration of vacancies
- through atomic disarrangements in solid solutions

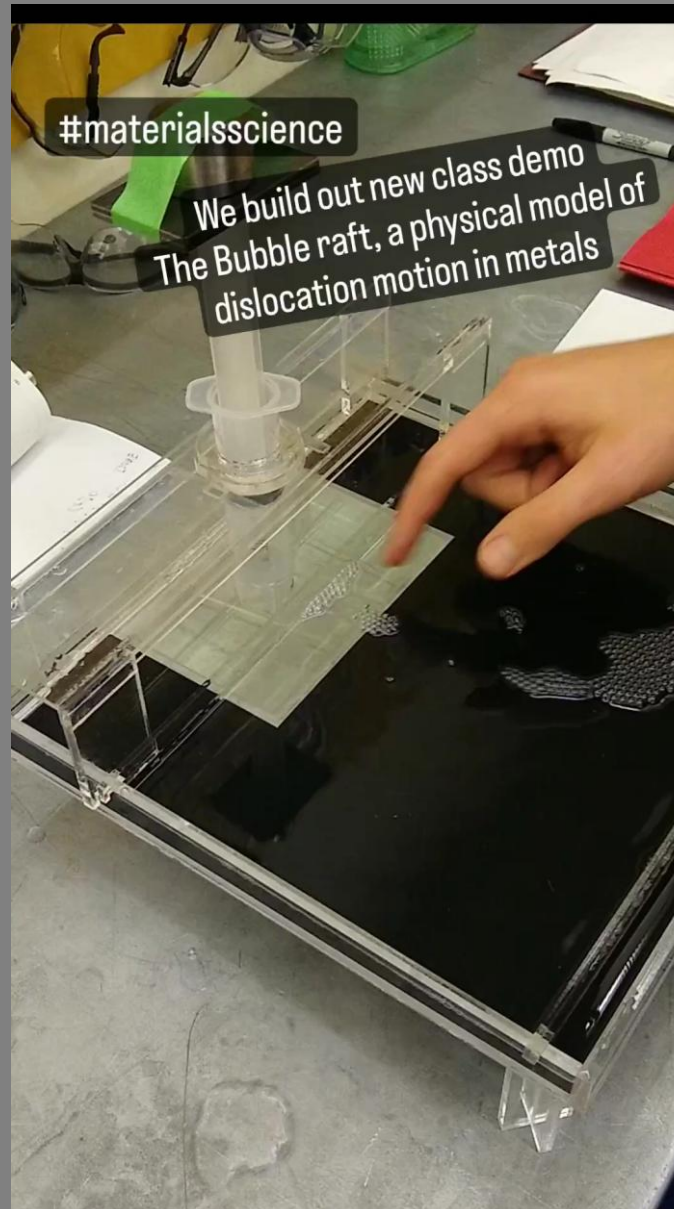
Types:

- edge dislocation (Taylor)
- screw dislocation (Burgers)
- mixed dislocation

LINEAR DEFECTS: DISLOCATIONS



LINEAR DEFECTS: DISLOCATIONS

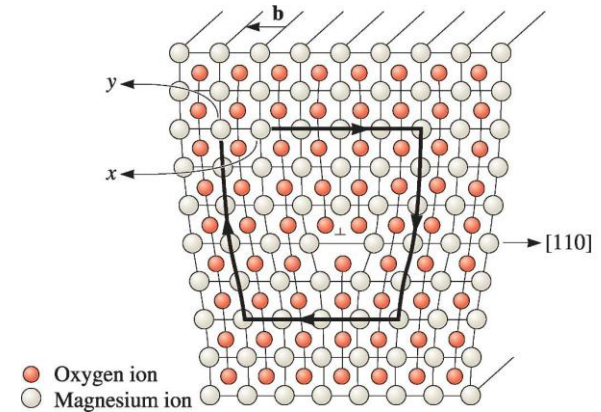
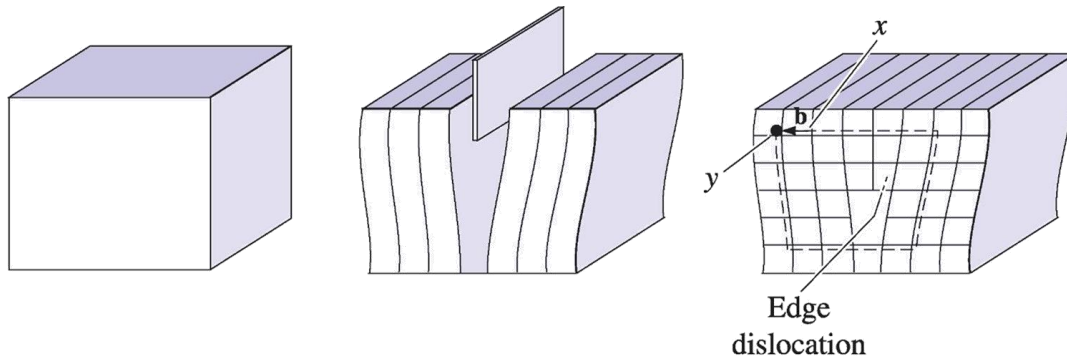


https://www.youtube.com/watch?v=3MH8bKDE_QU

LINEAR DEFECTS: EDGE DISLOCATIONS (or TAYLOR)

a) Edge dislocation (Taylor)

- ⇒ Geometric modification of the lattice: equivalent to an extra plane of atoms
- ⇒ Energetic modification (they store energy)



An edge dislocation in MgO showing the slip direction and Burger vector

Characterization of dislocation: Burgers vector ($\vec{b}$)

Burgers Vector: magnitude and direction of the lattice distortion associated with the dislocation*

*A burger vector $\mathbf{b}$ is required to close a loop of equal atom spacings around the edge dislocation

LINEAR DEFECTS: EDGE DISLOCATIONS (or TAYLOR)

a) Edge dislocation (Taylor)

Characterization of dislocation: Burgers vector ($\vec{b}$)

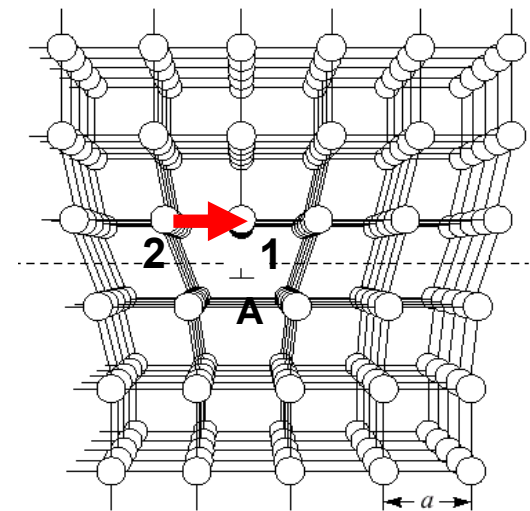
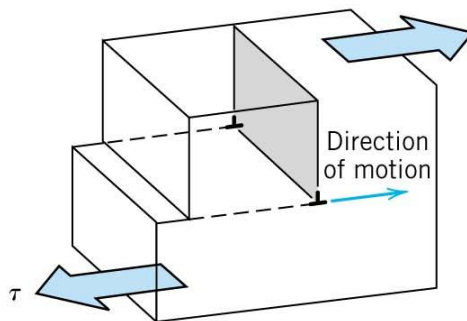
Magnitude: distance 1-2

Direction: 1-2 (or 2-1)

$b \perp$ to dislocation line

$b \parallel$ to the direction movement

$$E \propto |\vec{b}|^2$$

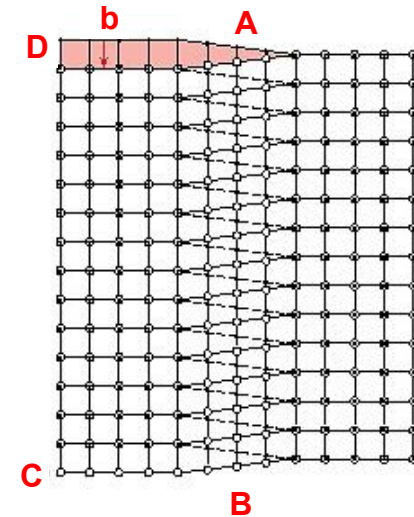
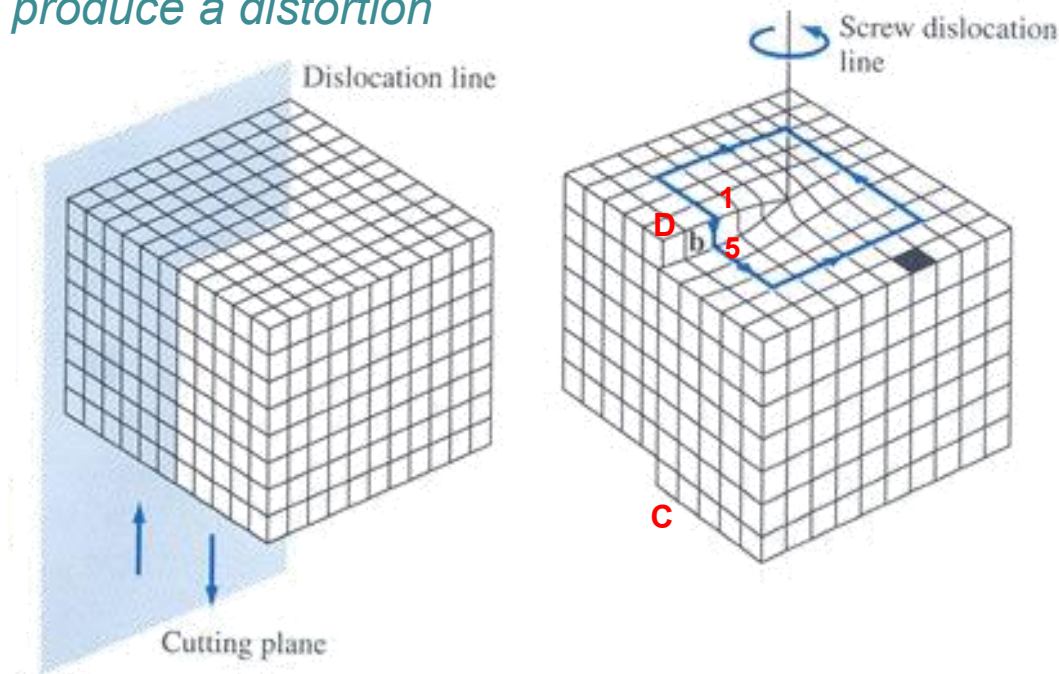


Dislocation line: $\perp$ point A
Slipping plane: line of points

LINEAR DEFECTS: SCREW DISLOCATIONS (or BURGER)

b) Screw dislocation (Burgers)

⇒ Locally curves some atom lines. *The effect is as if a shear stress is applied to produce a distortion*



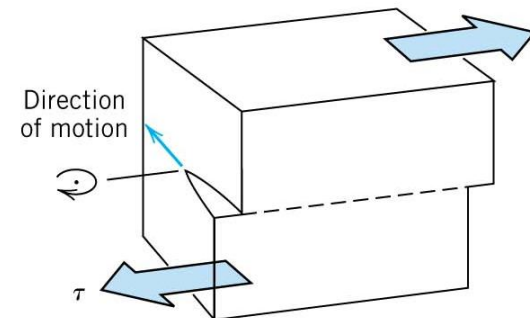
Plane view. We can trace a screw (spiral ramp) around the dislocation.

Burgers Vectors in screw dislocations:

The magnitude and distance from 1-5 define **b**

b || to dislocation line

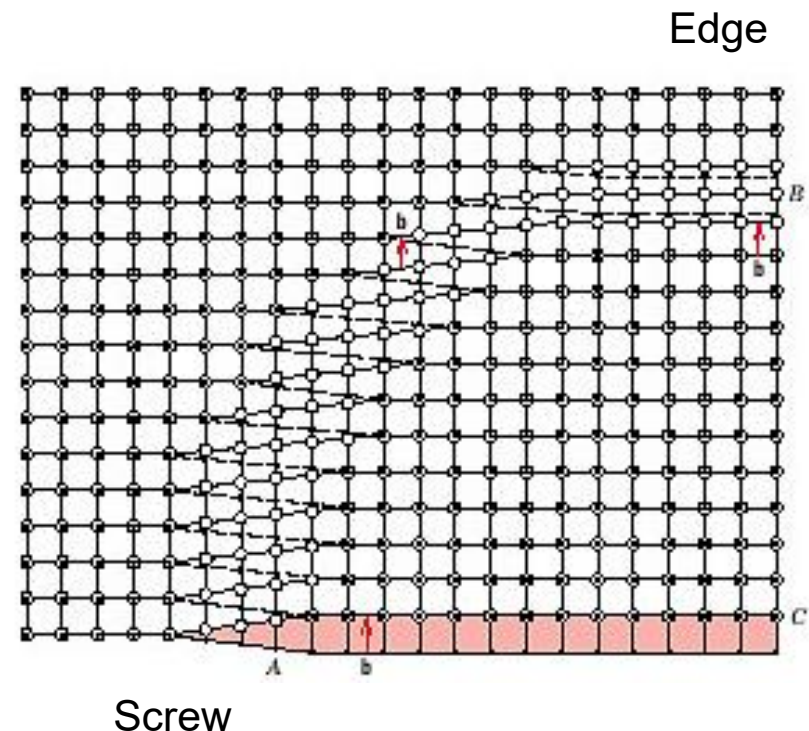
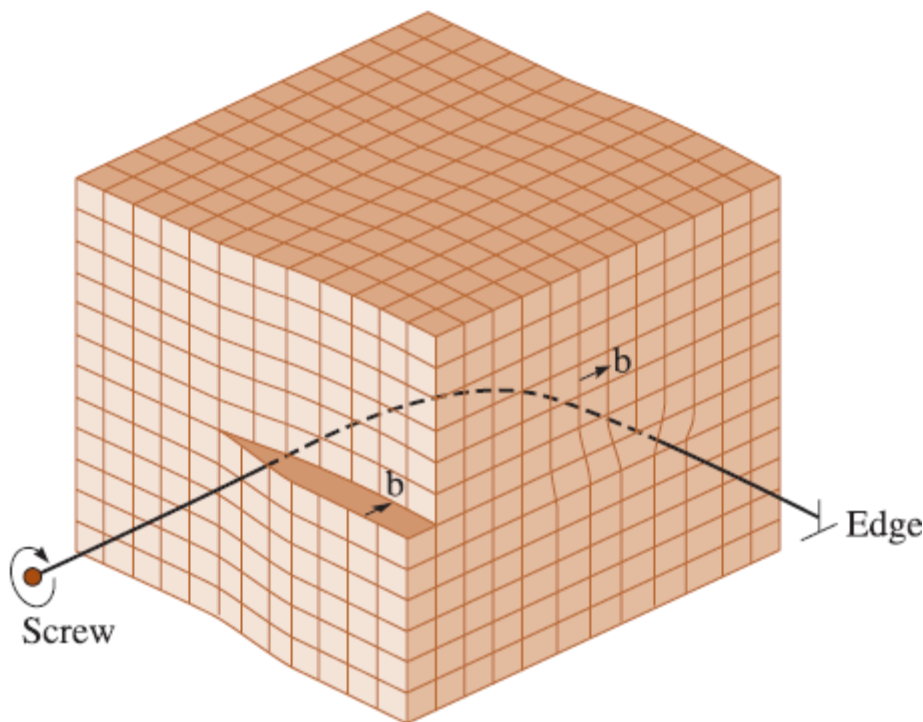
b ⊥ to the direction of movement



LINEAR DEFECTS: MIXED DISLOCATIONS

c) Mixed dislocation

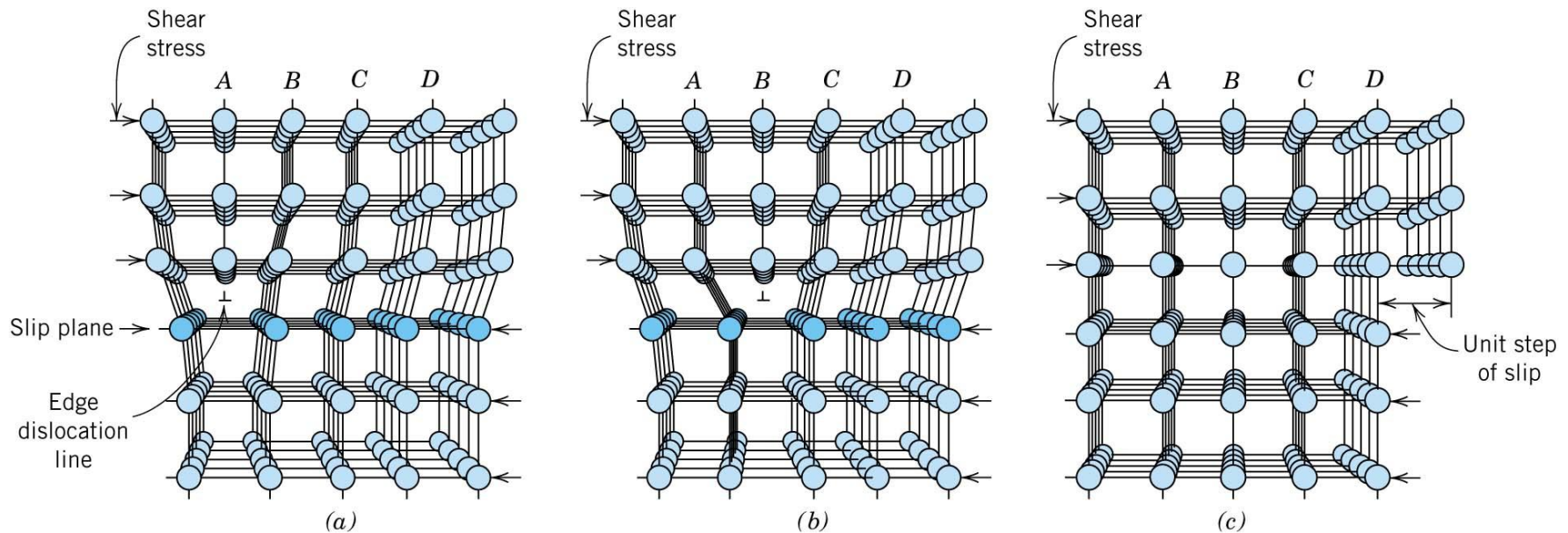
- ⇒ Edge and screw dislocations with a transition region between them.
- ⇒ Burgers vector remains the same for all portions of mixed dislocations.



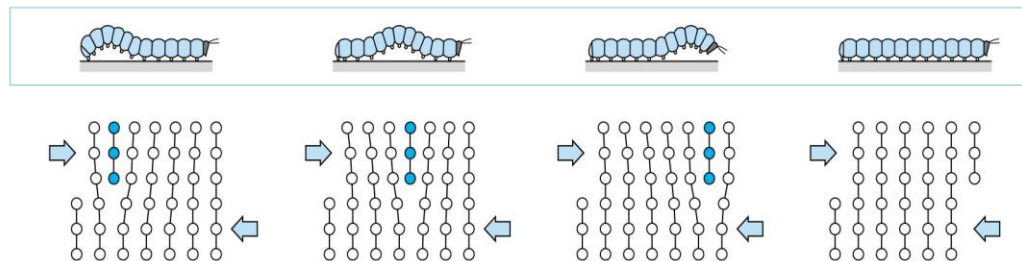
If there are many dislocations ⇒ they are mixed (**MIXED dislocations**)

DISLOCATIONS MOVEMENT

The capability of a metal to be plastically deformed depends on the capability of dislocations to move.

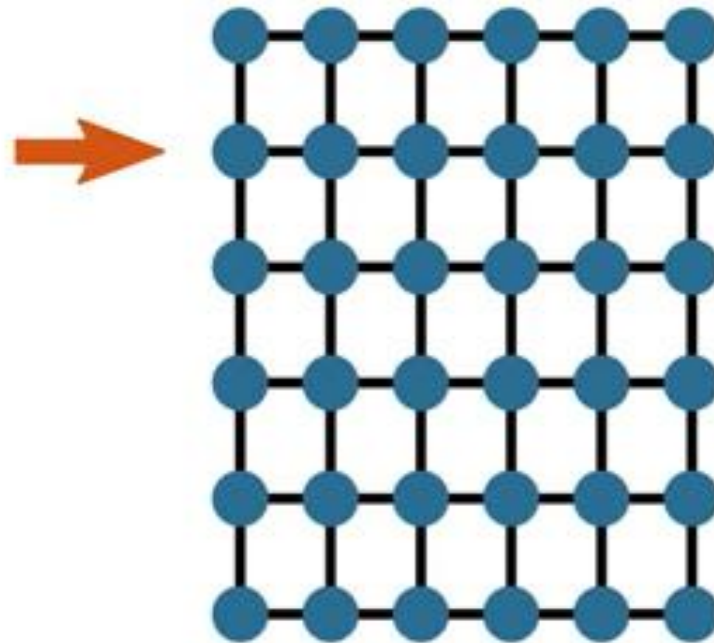


Dislocation movement (edge and screw) when a shear stress is applied



DISLOCATIONS MOVEMENT

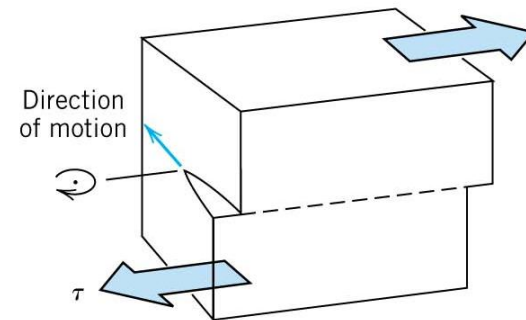
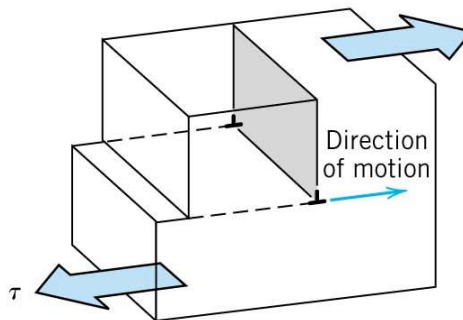
Strength and ductility in a metal depend on the capability of dislocations to move.



DISLOCATIONS MOVEMENT

Slip → process by which the plastic deformation is produced by dislocation motion

- Dislocation propagates along the **slip direction** over the **slip plane**
- Slip direction & slip plane form **slip system**
- For **edge** dislocations Burgers vector $\parallel$ to **slip direction**
- For **screw** dislocations Burgers vector $\perp$ to **slip direction**
- Slip direction is parallel to the direction of maximum packing



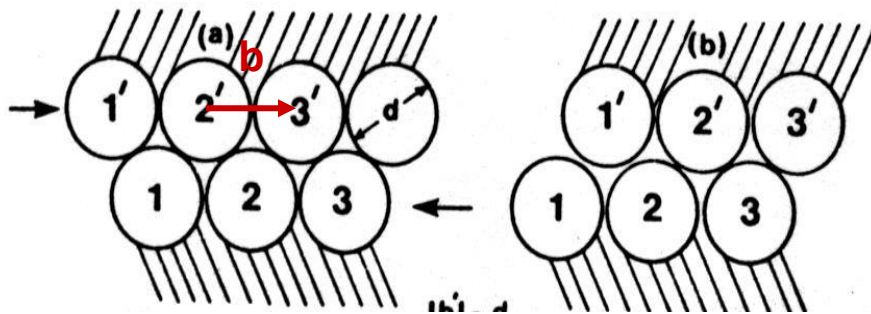
DISLOCATIONS MOVEMENT

Dislocation movement or slip direction is **parallel to the direction of maximum packing** since:

$$E \text{ used to move a dislocation} = E \propto |b|^2$$

Closed-packed direction

FCC $_{[110]}$

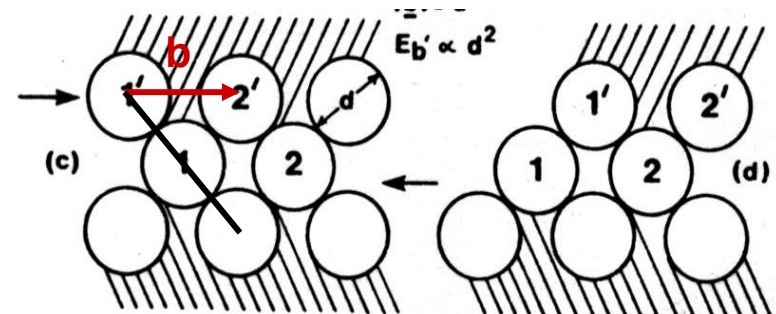


$$b = 2 \cdot R$$

$$E_b \propto 4R^2$$

Non closed-packed direction

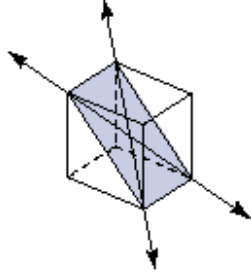
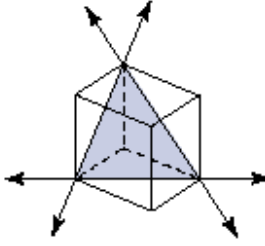
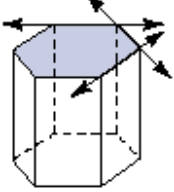
FCC $_{[100]}$



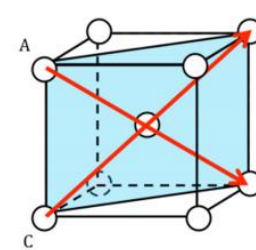
$$b^2 + b^2 = (4R)^2 \Rightarrow b = 2\sqrt{2} \cdot R$$

$$E_b \propto 8R^2$$

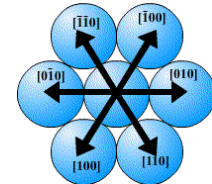
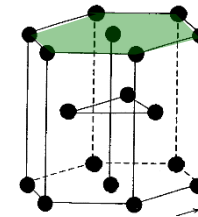
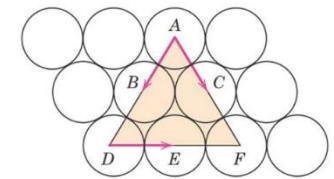
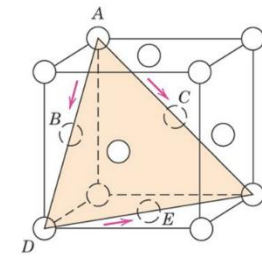
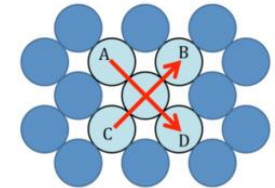
Slip planes and slip systems

Crystal structure	Slip plane	Slip direction	Number of slip systems	Unit cell geometry
bcc	{110}	$\langle \bar{1}11 \rangle$	$6 \times 2 = 12$	
fcc	{111}	$\langle \bar{1}\bar{1}0 \rangle$	$4 \times 3 = 12$	
hcp	(0001)	$\langle \bar{2}0 \rangle$	$1 \times 3 = 3$	

Most dense atomic packing



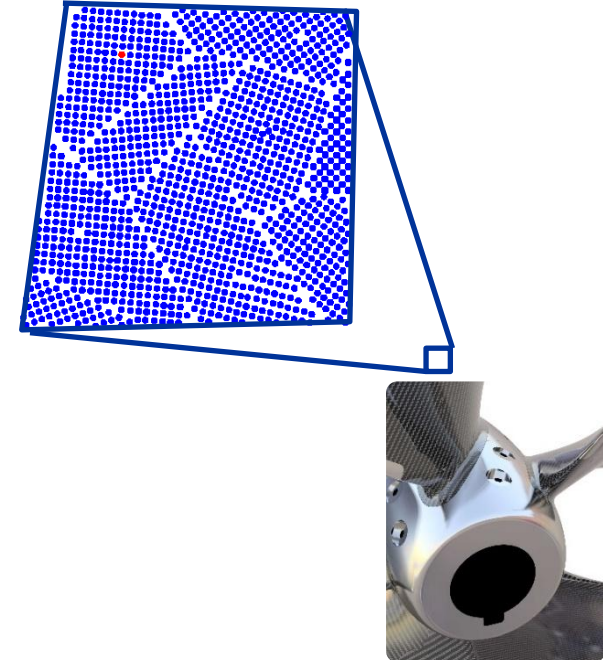
Higher linear density



↑ Slip systems → ↑ Ductility

TOPIC 2. STRUCTURE OF MATERIALS II

- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- Solid solutions
- Crystalline defects: linear defects
- **Crystalline defects: planar defects**
- Polymorphism and Allotropy



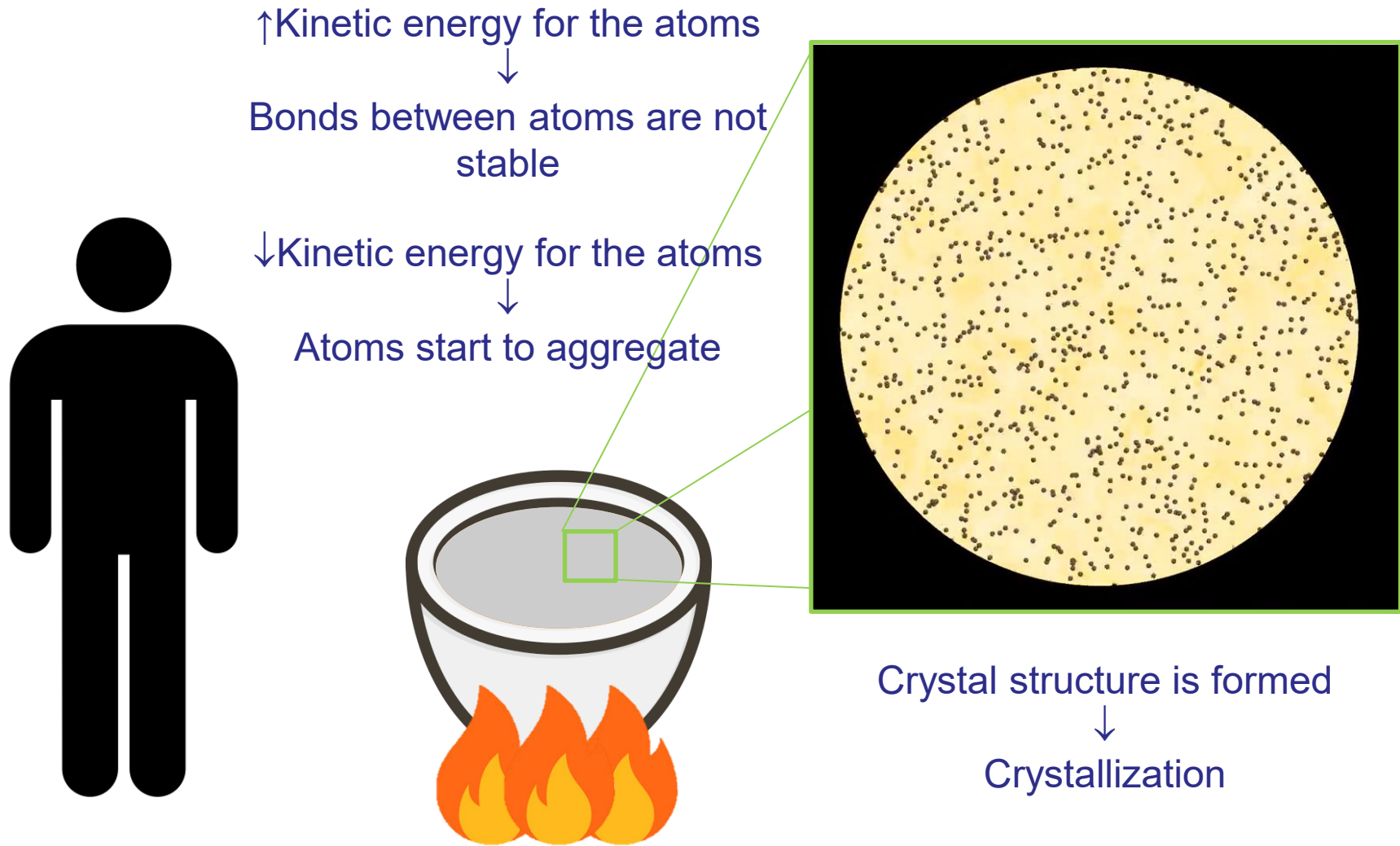
PLANAR DEFECTS

Planar Defects: Boundaries or planes that separate a material into regions, each having the same crystal structure but different orientation.

A) Grain boundaries

- 
- Grain boundaries
 - Twin boundaries
 - Stacking faults
 - External surfaces

GRAINS FORMATION



<https://www.youtube.com/watch?v=7xPfW2hYt8w>

PLANAR DEFECTS

Planar Defects: Boundaries or planes that separate a material into regions, each having the same crystal structure but different orientation.

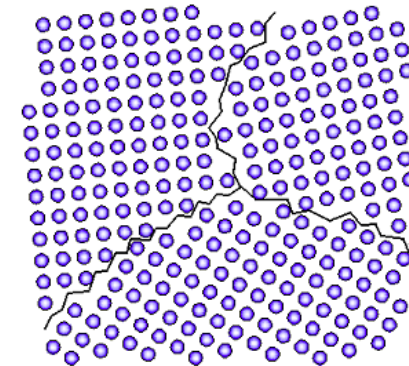
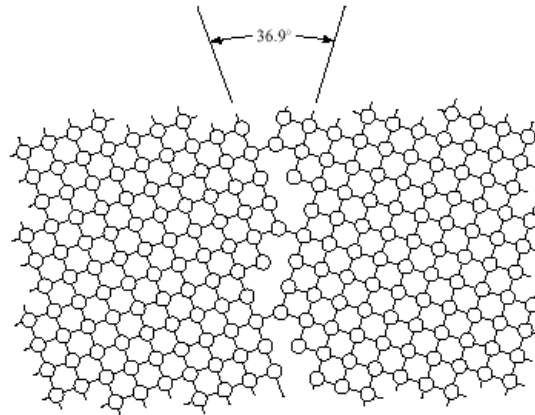
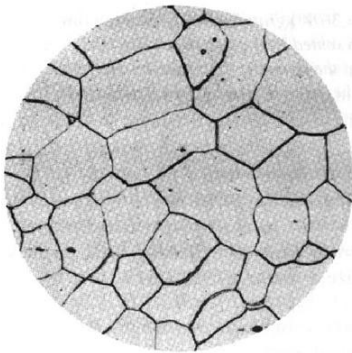
- Grain boundaries
- Twin boundaries
- Stacking faults
- External surfaces

A) Grain boundaries

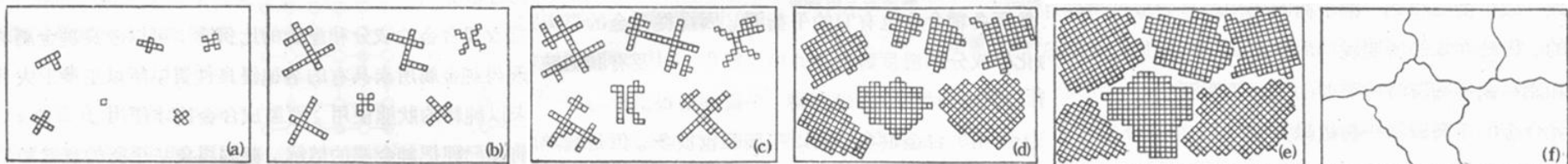
They separate crystals with different orientation.

Formation: during solidification: $\neq$ crystals are formed when $\neq$ nuclei grow simultaneously.

Polycrystal



Grains nucleation and growth



PLANAR DEFECTS: GRAIN BOUNDARIES

A) Grain boundaries

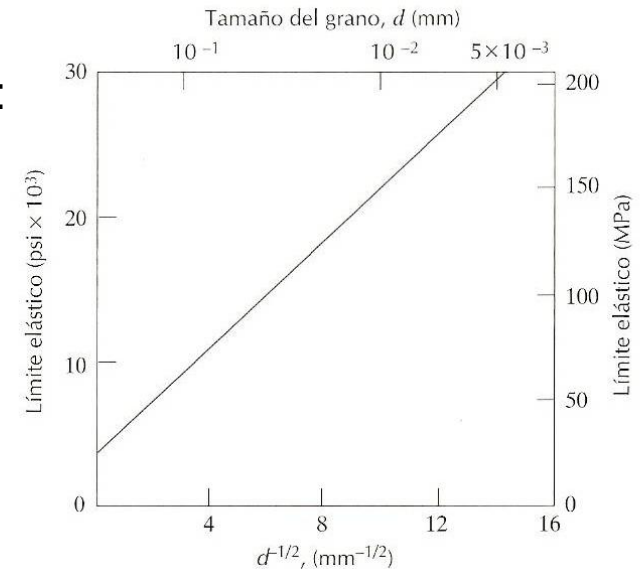
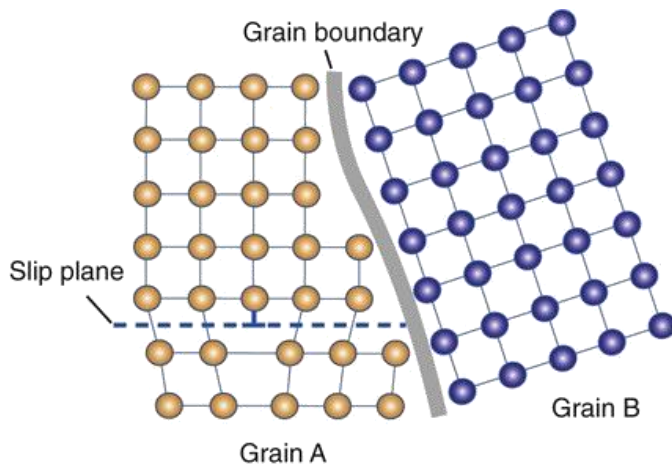
Boundary width: 2-5 interatomic distances

Very energetic zones (atomic packing < in interior):

- ✓ Solid state reactions
- ✓ ↑ Atomic diffusion

Materials properties = f (grain size)

⇒ Dislocation movement is limited when there are many grain boundaries



Grain size ↓ ⇒ Mechanical resistance ↑

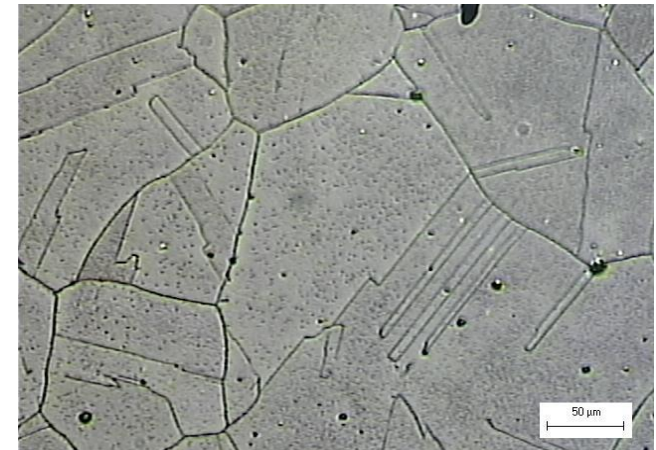
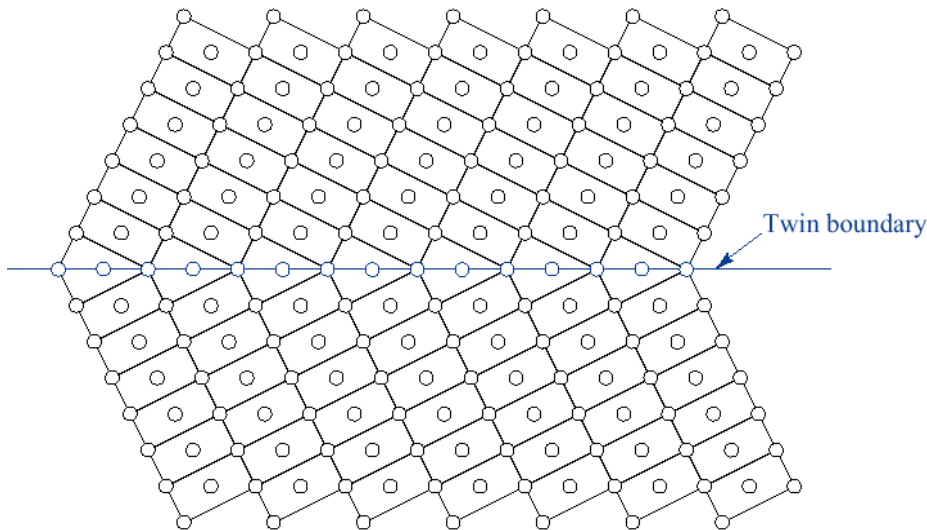
They limit dislocation movement in the material

PLANAR DEFECTS: TWIN BOUNDARIES

B) Twin boundary

Boundary that separates two parts of a crystal with a small difference in the orientation (misorientation) of the planes (crystal twinning).

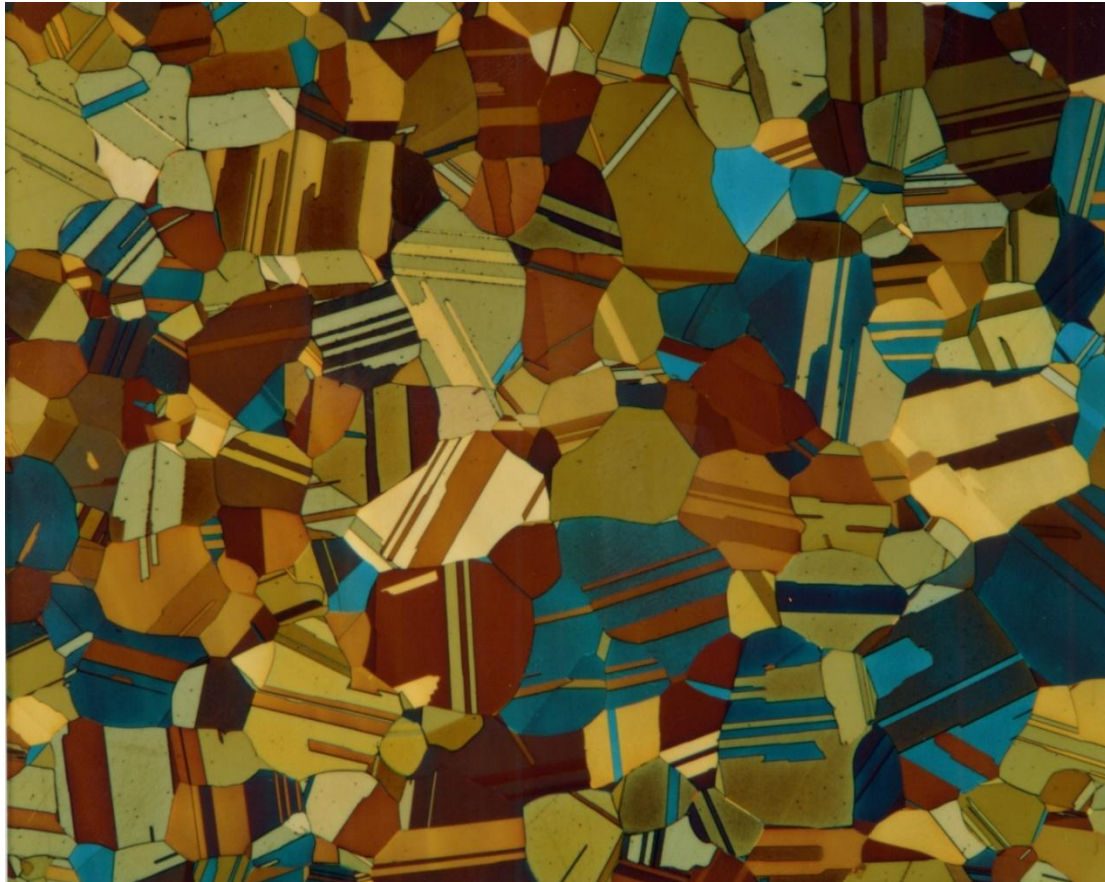
- Twin boundaries are grain boundaries with a special case of **mirror image misorientation** of the crystal structure, with a highly symmetrical interface
- Formation: deformation process or during heat treatments
- The **twin** boundaries interfere with the slip process and **increase mechanical strength** of the material $\Rightarrow$ make difficult dislocation slipping



Austenitic stainless steel

PLANAR DEFECTS: TWIN BOUNDARIES

B) Twin boundary



Brass (70 Cu / 30 Zn)

PLANAR DEFECTS: STACKING FAULTS

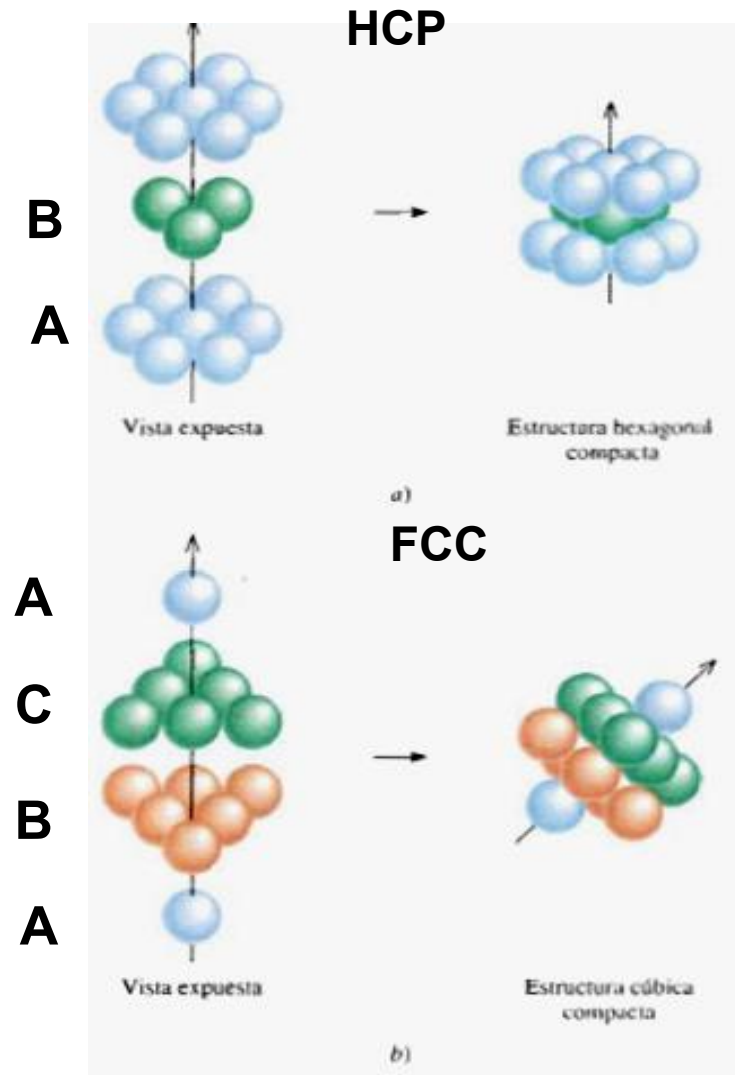
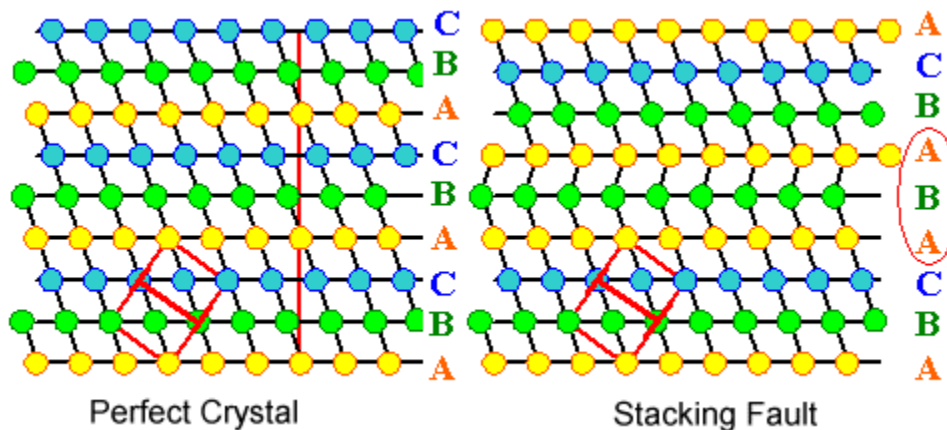
C) Stacking faults

A stacking fault is a one or two layer interruption in the **stacking sequence**

For example, FCC structure:

Perfect network: ABCABC ABC

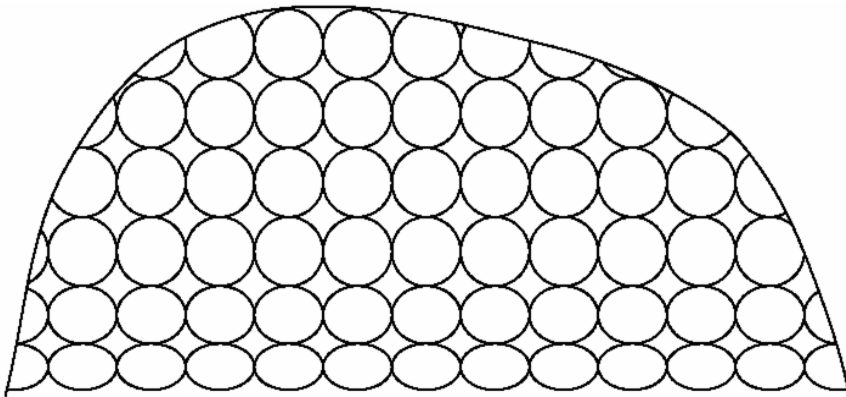
Stacking fault: ABCABABC



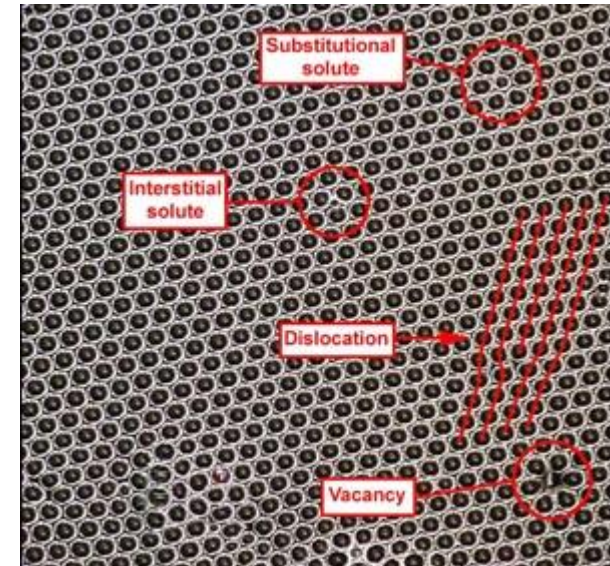
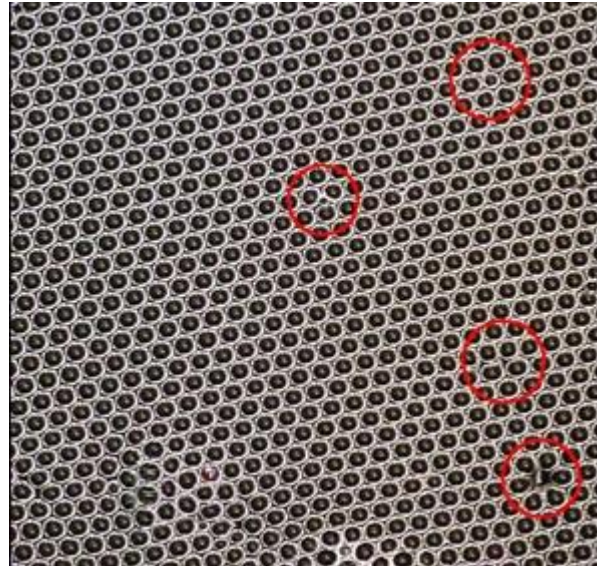
PLANAR DEFECTS: EXTERNAL SURFACES**D) External surface**

It is the end of the crystal or grain structure

Coordination numbers at the surface < at the interior of the crystal $\Rightarrow E_{\text{surface}} > E_{\text{interior}}$



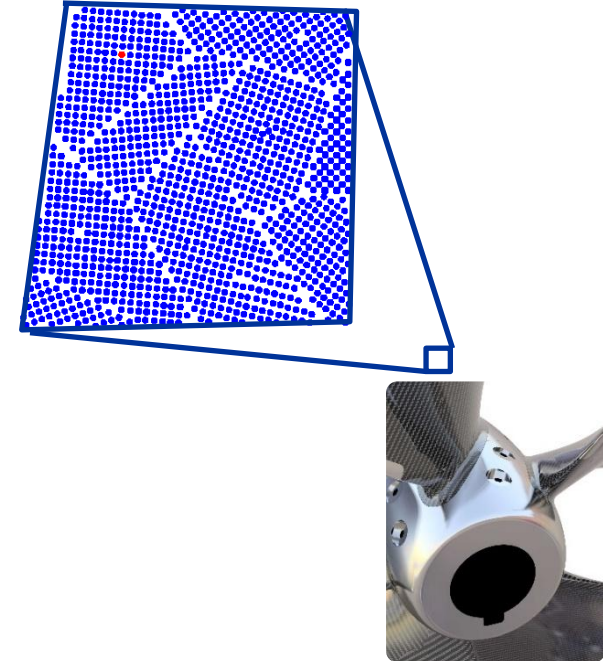
Bubble raft visualization of defects



- https://www.doitpoms.ac.uk/tlplib/dislocations/dislocation_glide.php/bubbleraft.php
- <https://www.youtube.com/watch?v=P345p7A4IW8>
- <https://www.youtube.com/watch?v=-qcys9XuNe8>

TOPIC 2. STRUCTURE OF MATERIALS II

- Calculation of the atomic, linear, planar density
- Crystalline defects: point defects
- Solid solutions
- Crystalline defects: linear defects
- Crystalline defects: planar defects
- **Polymorphism and Allotropy**



POLYMORPHISM AND ALLOTROPY

Polymorphism: the ability of a solid material to exist in more than one crystal structure.

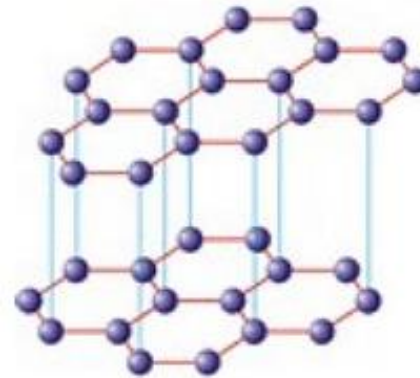
Allotropy: the ability of *chemical elements* to exist in two or more different forms in the same physical state

⇒ The same element or compound can exist in more than one crystalline state under $\neq$ conditions of P and T.

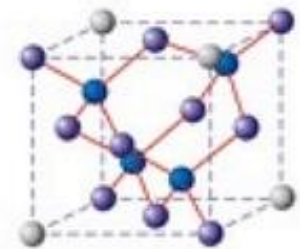
Examples

Allotropes of Carbon

- Diamond ⇒ 3D covalent structure ⇒ hard, transparent, insulator
- Graphite ⇒ Laminar structure ⇒ Secondary bonds between layers ⇒ soft, non transparent, conductor



graphite



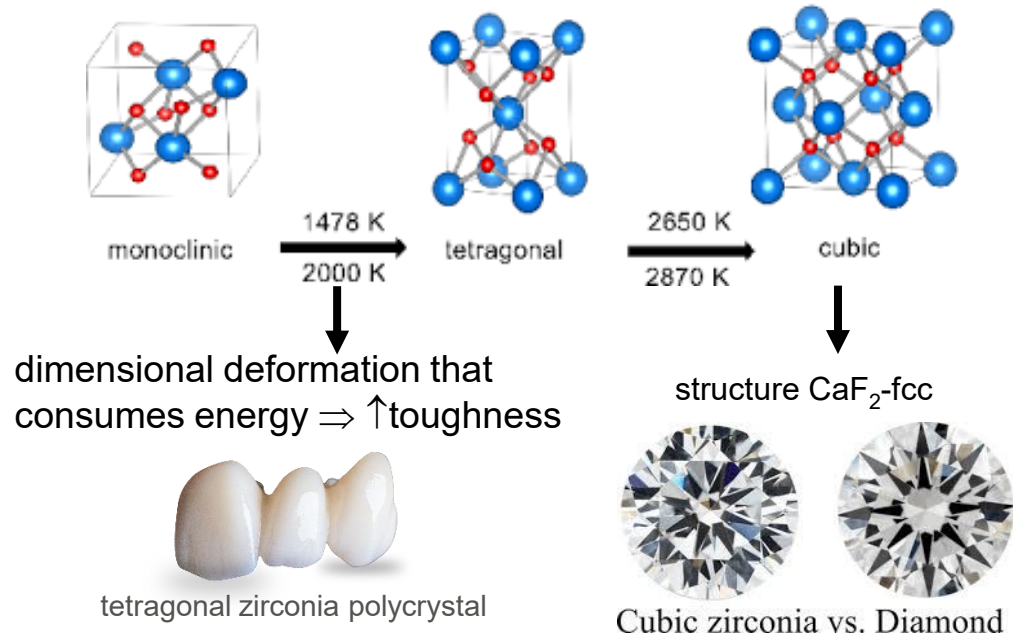
diamond



POLYMORPHISM AND ALLOTROPY

Examples

Zirconia (ZrO_2)



Iron (steels):

BCC at room temperature; FCC from 910 °C $\Rightarrow$ possibility of heat treatment

